Fiscal Year:	FY 2009	**Task Last Updated:**	FY 08/12/2009
PI Name:	Crum, Lawrence A. Ph.D.		
Project Title:	Smart Therapeutic Ultrasound Device for Mission-Critical Medical Care		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Smart Medical Systems and Technology Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **ExMC**:Exploration Medical Capabilities		
Human Research Program Risks:	(1) **Medical Conditions**:Risk of Adverse Health Outcomes and Decrements in Performance Due to Medical Conditions that occur in Mission, as well as Long Term Health Outcomes Due to Mission Exposures		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	lac@apl.washington.edu	**Fax:**	FY 206-543-3702
PI Organization Type:	UNIVERSITY	**Phone:**	206-685-8622
Organization Name:	University of Washington		
PI Address 1:	Applied Physics Laboratory		
PI Address 2:	1013 N.E. 40th Street		
PI Web Page:			
City:	Seattle	**State:**	WA
Zip Code:	98105-6606	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
Start Date:	08/01/2008	**End Date:**	07/31/2012
No. of Post Docs:	3	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bailey, Michael (University of Washington) Sapozhnikov, Oleg (University of Washington) Carter, Stephen (University of Washington)		
Grant/Contract No.:	NCC 9-58-SMST01601		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The major goals of this effort is to utilize existing ultrasound platforms and the concept of image-guided therapy to control traumatic bleeding, ablate cancerous tumors, and diagnose and clear kidney stones. Our methods and devices are countermeasures to specific risks described in the Human Research Program Integrated Research Plan, viz., (1) Lack of advanced therapeutic capability, (2) lack of capability to treat renal stones, and (3) lack of non-invasive diagnostic imaging capabilities. The original specific aims are: Specific Aim 1: Support ongoing leveraged efforts in Acoustic Hemostasis (AH) and HIFU Tumor Ablation (TA) by addressing fundamental scientific issues as well as to ensure NSBRI relevance. Specific Aim 2: Develop methods and technology that would enable detection of renal stones with ultrasound. Specific Aim 3: Develop technology and perform in vitro studies of stone comminution. Specific Aim 4: Utilizing technology and protocols developed in SAs 2 and 3, perform in vivo studies in a porcine model. (2) The key findings and associated research productivity for year 1 are: - Developed an automated ultrasound guided high intensity focused ultrasound (HIFU) system to detect and stop bleeding: Published paper. - Began initial investigation of HIFU induced tumor specific immune response in collaboration with Fred Hutchinson Cancer Research Institute: Obtained NIH funding to support a Postdoctoral fellow on the work. - Developed a body of evidence on methods to accelerate HIFU therapy with the use of shock waves: Papers published. - Tested new Doppler ultrasound-based kidney stone detection method in vitro, in animals, and in humans: Filed 3 Records of Invention with the University of Washington TechTransfer Office. - Developed method to use focused ultrasound to move kidney stones and stone fragments within the kidney to expedite stone clearance: Presented work to American Urology Association. - Miniaturized device to size stone fragments for safe extraction and tested operation in kidney: Submitted U.S. utility patent application and published paper. - Developed correlation between ultrasound-induced and monitored vasoconstriction; discovered that vasoconstriction reduces injury during stone fragmentation therapy: Published paper. - Participated in the generation of a white paper through gap analysis of medical risk 4.15 (Lack of lithotripsy or other capability to treat a renal stone) by the JSC Exploration Medical Capability element of the Human Research Program. - Copyrighted and licensed technology describing new HIFU sources and test equipment. (3) These findings are self-explanatory, but we wish to highlight the broader impact on kidney stone disease. Our new detection technique requires only a software change, at most, to existing ultrasound technology on board ISS. It is sensitive and easy to use. Our belief is that this approach will provide NASA with the capability to detect even small, asymptomatic stones. The next phase of our work will be to use focused ultrasound (which could also be generated with only modest software reprogramming of the existing ISS ultrasound device) to dislodge the stone and push it toward the opening of the ureter where it could be naturally passed. In this way, a potential critical clinical problem would be solved by early diagnosis and prevention, rather than by last-minute and difficult therapy. This methodology has obvious and significant earthbound utility as well. (4) Our future plans will focus on continued automation, cancer treatment, and the prevention of complications from kidney stones. We have obtained access to a programmable ultrasound imager that we will program to test our new stone detection algorithms. The immune response study, initiated by seed funds from our NSBRI cost match, will be continued under NIH sponsorship; hopefully, it will be determined that HIFU can induce a systemic tumor-specific immune response in mice. Investigation and improvement of the stone detection technique will continue by direct comparison of simulation and measurement. The technique to detect stones at pre-symptomatic levels will be tested against standard ultrasound, fluoroscopy, and CT in patients. Acceleration of stone passage by focused ultrasound will be investigated in a porcine animal model.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:	In general, this project advances a new technology- high intensity focused ultrasound (HIFU) toward clinical application. HIFU machines have treated over 100,000 cancer patients in China and over 5,000 in Europe and Japan. The only indication approved by the FDA for the use of HIFU on patients is that of uterine fibroids. Dozens of start-up companies and the big three ultrasound companies - GE, Philips, and Siemens - are developing HIFU machines. We are beginning to license our intellectual property to the commercial sector. We were intimately involved in the founding of the International Society for Therapeutic Ultrasound and currently play major roles in its administration. Similarly, we have helped establish the International Kidney Stone Institute and have authored consensus reports on lithotripsy and ultrasound safety. Specifically our work this year has provided the following Earthbound benefits, viz., 1. We demonstrated a way to automate the detection and treatment of bleeding. 2. We have designed and initiated a study to test the hypothesis that HIFU can generate a systemic immune response and have high hopes for progress in this area. 3. We have offered the HIFU community significant insight into how to plan, execute and monitor HIFU treatments. For earthbound HIFU, we have raised considerable concern over the accuracy of the gold standard (MR thermometry) used to "ensure" heating only where desired.

	4. Our efforts to develop a Doppler ultrasound-based, kidney stone detection method has several applications. It appears at least as good a fluoroscopy in targeting and could therefore replace this approach and its ionizing radiation. It also can be used real-time and therefore could compensate for respiratory motion during treatment. Lastly, an accurate ultrasound imaging system could be used in the urologist's office to localize stones and to replace the need for CT scans. 5. Our new method to use focused ultrasound to move kidney stones could be used whenever residual stones are observed after treatment. These stones get trapped and do not pass naturally. They then serve as a nucleus for future stones. 6. Our miniaturized device to size stone fragments may soon be used during ureteroscopy to determine stone size before attempting to extract stones too large to pass through a finite-sized lumen. 7. We have licensed new HIFU sources and test equipment to a vendor who will provide these tools to researchers, clinicians, regulators, and manufacturers to accelerate the implementation of clinical HIFU applications.
Task Progress:	The second tier research tasks described in the proposal are listed below and the progress on them for this year is as follows. Task 1A1. Perform studies of bleeding detection in a flow-phantom model. A model and a method to excise but not detach the femoral arteries in live pigs have been developed. Task 1B1. Perform studies to determine pressure and temperature in ex vivo tissue exposed to HIFU. A paper co-authored by M.S. Canney, V. A. Khokhlova, O.V. Bessonova, M. R. Bailey, and L. A. Crum, entitled "Millisecond boiling produced by high intensity focused ultrasound," was submitted in February 2009 to the journal Ultrasound in Medicine and Biology. Task 2A. Develop new stone detection techniques based on radiation force and reverberation responsible for the twinkling artifact and vibroacoustography. We have completed considerable data collection and have begun an analysis of the data to improve understanding of the mechanisms and algorithms for use. Three records of invention describing the evidence for this new understanding have been filed. We also reported progress in the following paper: A. Shah, M. Paun, J. Kucewicz, O. A. Sapozhnikov, M. Dighe, H. A. McKay, M. D. Sorensen, and M. R. Bailey, "Investigation of an ultrasound imaging technique to target kidney stones in lithotripsy," J. Acoust. Soc. Am., 125(4, Pt. 2), 2620 (2009). Task 2B. Investigate stone-sizing technology in tissue. We have submitted for publication our progress in this task in the following article: M.D. Sorensen, J.M.H. Teichman, and M.R. Bailey, "A Proof of Principle of a Prototype Ultrasound Technology to Size Stone Fragments During Ureteroscopy," J. Endourology in press 2009, We have also filed U.S. and international utility patent applications, and are negotiating licensing. Task 3A. Utilize the YUANDE HIFU tumor ablation device as a platform for determining the acoustic protocols necessary for ultrasound-based stone comminution. No significant progress to report. For business reasons, effort has shifted to a second clinical device. Task 3B. Engineer and optimize an image-guided, two-frequency HIFU system for renal stone comminution. In year one, we have focused on moving small stones within the kidney with ultrasound to facilitate natural stone clearance. Results were reported at the American Urology and Laparoendoscopic Surgeons meetings. Stone comminution work is scheduled to continue later in the Research Project. Task 4A. Perform in vivo tests of the imaging protocols developed in Task 2. Investigations are underway for stone detection not only in animals but in humans. Human protocols have been approved and added to the grant. Task 4B. Perform studies to determine the potential for HIFU-induced stone comminution as well as any associated tissue injury. In vivo studies of stone clearance are underway.
Bibliography Type:	Description: (Last Updated: 03/22/2019)
Articles in Peer-reviewed Journals	Bessonova OV, Khokhlova VA, Bailey MR, Canney MS, Crum LA. "Focusing of high intensity ultrasound beams and ultimate values of shock wave parameters." Acoustical Physics. In press, June 2009. , Jun-2009
Articles in Peer-reviewed Journals	Canney MS, Bailey MR, Crum LA, Khokhlova VA, Sapozhnikov OA. "Acoustic characterization of high intensity focused ultrasound fields: a combined measurement and modeling approach." J Acoust Soc Am. 2008 Oct;124(4):2406-20. PMID: 19062878 , Oct-2008
Articles in Peer-reviewed Journals	Khokhlova TD, Canney MS, Lee D, Marro KI, Crum LA, Khokhlova VA, Bailey MR. "Magnetic resonance imaging of boiling induced by high intensity focused ultrasound." J Acoust Soc Am. 2009 Apr;125(4):2420-31. PMID: 19354416 , Apr-2009
Articles in Peer-reviewed Journals	Kreider W, Bailey MR, Ketterling JA. "Beamwidth measurement of individual lithotripter shock waves." J Acoust Soc Am. 2009 Feb;125(2):1240-5. PMID: 19206897 , Feb-2009
Articles in Peer-reviewed Journals	Kucewicz JC, Bailey MR, Kaczkowski PJ, Carter SJ. "Focused ultrasound: concept for automated transcutaneous control of hemorrhage in austere settings." Aviat Space Environ Med. 2009 Apr;80(4):391-4. PMID: 19378911 , Apr-2009
Articles in Peer-reviewed Journals	Sorensen MD, Shah AR, Canney MS, Sapozhnikov OA, Teichman TM H, Bailey MR. "Ureteroscopic ultrasound technology to size kidney stone fragments: Proof of principle using a miniaturized probe in a porcine model." J Endourol. Submitted, June 2009. , Jun-2009
Articles in Peer-reviewed Journals	Sorensen MD, Teichman JM, Bailey MR. "Proof of principle in vitro study of a prototype ultrasound technology to size stone fragments during ureteroscopy." J Endourol. 2009 Jul;23(7):1161-4. PubMed PMID: 19538061 , Jul-2009
Articles in Peer-reviewed Journals	Canney MS, Khokhlova VA, Bessonova OV, Bailey MR, Crum LA. "Millisecond boiling produced by high intensity focused ultrasound." Ultrasound in Medicine & Biology, Submitted February 2009. , Feb-2009

Awards	Bailey MR. "Re-elected AIUM Bio-effects committee, February 2009." Feb-2009
Awards	Crum LA. "Invited tutorial lecture at 2009 IEEE Ultrasonics Symposium, September 2009." Sep-2009
Awards	Khokhlova TD. "NIH T32 Training grant postdoctoral fellowship, July 2009." Jul-2009
Awards	Bailey M. "Elected to the Governing Board of the Interntional Society for Therapeutic Ultrasound, June 2009." Jun-2009
Awards	Sapozhnikov OA. "Fellow of the Acoustical Society of America, June 2009." Jun-2009
Dissertations and Theses	Canney MS. "Nonlinear enhancement of heating due to shock formation in high intensity focused ultrasound fields." Dissertation, University of Washington, June 2009. , Jun-2009
Papers from Meeting Proceedings	Crum L, Bailey M, Canney M, Chen H, Matula T, Mc Innes C. "Direct observation of surface particle removal by an ultrasonic toothbrush." Acoustics '08 Paris, Joint meeting of the Acoustical Society of America, ASA, the European Acoustics Association, EAA, and the Société Française d'Acoustique, SFA, Paris, France, June 29-July 4, 2008. Proceedings of Acoustics '08 Paris, Joint meeting of the Acoustical Society of America, ASA, the European Acoustics Association, EAA, and the Société Française d'Acoustique, SFA, Paris, France, June 29-July 4, 2008. , Jul-2008
Papers from Meeting Proceedings	Maxwell AD, Sapozhnikov OA, Pishchalnikov YA, Bailey MR. "Simulation and measurement of PVDF membrane hydrophone low-frequency response for accurate measurements of lithotripsy shockwaves." 2008 IEEE Ultrasonics Symposium, Beijing, China, November 2-5, 2008. Proceedings of 2008 IEEE Ultrasonics Symposium, November 2008. , Nov-2008
Papers from Meeting Proceedings	Pishchalnikov YA, McAteer JA, Pishchalnikova IV, Williams JC Jr, Bailey MR, Sapozhnikov OA. "Bubble proliferation in shock wave lithotripsy occurs during inertial collapse." 18th International Symposium on Nonlinear Acoustics, Stockholm, Sweden, July 7-10, 2008. Proceedings from 18th International Symposium on Nonlinear Acoustics, July 2008. , Jul-2008
Papers from Meeting Proceedings	Sorensen MD, Teichman JM H, Bailey MR. "A prototype ultrasound instrument to size stone fragments during ureteroscopy." 2nd International Urolithiasis Research Symposium, Indianapolis, IN, April 17-18, 2008. Proceedings of 2nd International Urolithiasis Research Symposium. AIP Conference Proceedings, vol. 1049, p. 348-352, September 18, 2008. http://dx.doi.org/10.1063/1.2998057 , Sep-2008
Papers from Meeting Proceedings	Shah A, Paun M, Kucewicz J, Sapozhnikov OA, Dighe M, McKay HA, Sorensen MD, Bailey MR, Crum LA. "Invited paper: Investigation of an ultrasound imaging technique to target kidney stones." 157th Meeting of the Acoustical Society of America, Portland, Oregon, May 18-22 2009. J Acoust Soc Am. 2009 Apr;125(4, Pt 2):2620. , Apr-2009
Patents	111. Patent, June 2008. Jun-2008 Bailey MB, Carter S, Kaczkowski P, Kucewicz J. "Method and system for automation of focused ultrasound treatment."
Patents	None yet. Patent in process, May 2008. May-2008 Bailey MB, Sorensen MD, Teichman J. "Method and device to measure stone size in surgical removal of kidney stones."
Significant Media Coverage	Evangelista B. "Newspaper article: Some 'Star Trek' gadgets no longer futuristic." San Francisco Chronicle newspaper, Sunday, May 10, 2009., May-2009
Significant Media Coverage	Shah A, Paun M, Kucewicz J, Sapozhnikov OA, Dighe M, McKay HA, Sorensen MD, Bailey MR, Crum LA. "Lay version of paper for Meeting Press: Investigation of an ultrasound imaging technique to target kidney stones." Acoustical Society of America lay paper, May 2009. http://acoustics.org/pressroom/httpdocs/157th/bailey.html , May-2009